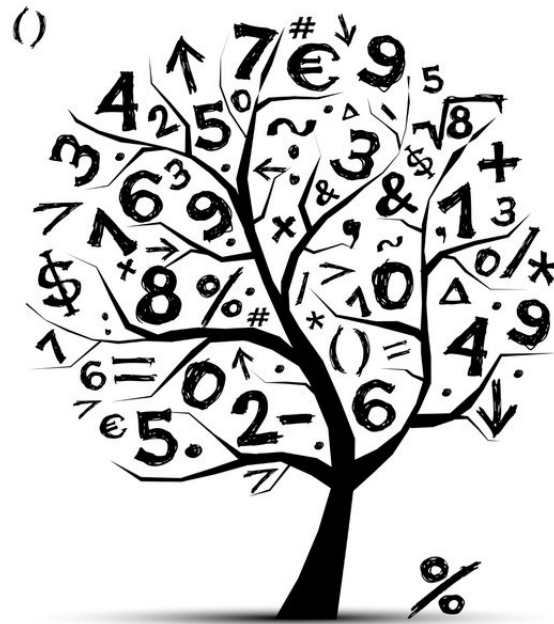


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Modular arithmetic



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1 Euclid's division

Lemma 1: Euclid's division (or division with remainder)

For every pair of integer numbers n and m , with $m \neq 0$, there exist unique integer numbers q and r such that

$$n = qm + r \quad \text{and} \quad 0 \leq r < |m| \quad (1)$$

- The value n is called the **dividend**, the value m is called the **divisor**, the value q is called the **quotient** and the value r is called the **remainder**.
- The standard convention is that the remainder is non negative but there are other conventions where the remainder r can be negative.

Proof.

- To prove *existence*, we consider the set

$$S = \{ n - km : k \in \mathbb{Z} \text{ and } n - km \geq 0 \}$$

The set S is nonempty, since for $k = -|n|m$ we have $n - km = n + |n|m^2 \geq n + |n| \geq 0$. Since S is a nonempty subset of the nonnegative integers, the well-ordering principle guarantees that there exists a smallest element $r \in S$, meaning there exists some integer q such that

$$r = n - qm, \quad \text{with } r \geq 0$$

By construction, we must have $r < |m|$, since if $r \geq |m|$, then we could write

$$r' = r - |m| = n - \left(q + \frac{|m|}{m} \right) m \geq 0$$

which would imply $r' \in S$, contradicting the minimality of r .

- To prove *uniqueness*, suppose there exist two pairs (q_1, r_1) and (q_2, r_2) such that

$$n = q_1 m + r_1 = q_2 m + r_2$$

with $0 \leq r_1, r_2 < |m|$. Subtracting the two equations, we have

$$(q_1 - q_2)m = r_2 - r_1$$

Since $|r_2 - r_1| < |m|$ and m divides $r_2 - r_1$ (which equals the left-hand side), the only possibility is $r_1 = r_2$, which implies $q_1 = q_2$, ensuring uniqueness.

□

Definition 1: multiple

An integer number n is a **multiple** of an integer number m if there exists an integer number q such that

$$n = q \cdot m \quad (2)$$

Definition 2: factor

An integer number m is a **factor** of an integer number n if there exists an integer number q such that

$$n = q \cdot m \quad (3)$$

- An integer number n is **divisible** by an integer number m if m is a factor of n .

2 Modulo and modulo congruence

Given two integer numbers n and m , with $m \neq 0$, n **modulo** m (abbreviated in $n \bmod m$) is the **remainder** r of the Euclidean division (or division with remainder) of n by m , where n is the **dividend** and m is the **divisor**.

- It follows from the definition that:

$$0 \leq n \bmod m < |m| \quad \forall n, m \in \mathbb{Z}, m \neq 0$$

Two natural numbers n and k are **congruent** modulo a natural number $m \geq 1$ if

$$n \bmod m = k \bmod m$$

and we write

$$n \equiv k \pmod{m}$$

- The parentheses mean that $(\bmod m)$ applies to the entire equation, not just to the right-hand side (here, k).
- Equivalently, we have $n \equiv k \pmod{m}$
 1. if n and k have the same remainder when divided by m .
 2. if m is a divisor of $|n - k|$, i.e., if there exists a natural number q such that $|n - k| = q \cdot m$ or equivalently if there exists an integer number q such that $n - k = q \cdot m$

Example 1: congruence modulo 5

For example, 23 and 13 are congruent modulo 5 and we have $23 \equiv 13 \pmod{5}$